

Jonathan Salfity

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Austin, TX | U.S. Citizen

SUMMARY

- Final year PhD student with 7 years of corporate and 4 years of startup experience.
- Research focus in robotic skill & behavior composition, modular robots, control theory, GenAI.
- Corporate and startup experience in team management, fundraising, finance, and GTM.

ACADEMIC

Ph.D., Mechanical Engineering, *UT Austin*, Robotics and AI (Expected) 2026
Advisor: Mitch Pryor
Group: Nuclear and Applied Robotics Group
M.S., Mechanical Engineering, *UCLA*, Systems & Control, Robotics 2014
B.S., Mechanical Engineering, *UCLA* 2013

RESEARCH FOCUS

- Skill & Behavior Composition for Modular Robotics
- GenAI, Foundation Models for Robotic Behaviors and Skills
- Robust and Optimal Control for Nonlinear Systems

PROFESSIONAL

geCKo Materials Campbell, CA | 2021 - Current
Co-Founder / Advisor / Board Member

- Co-founded geCKo Materials to bring biomimicry, industrial grade adhesives to market.
- Contributed to \$10M raise, expanded the team to 10 FTEs operating in a 10,000 sq. ft. manufacturing facility, ensuring weekly product shipments.

General Motors Mountain View, CA / Austin, TX | Summer 2026
Summer Researcher

- Built a configurable scripted-policy API to generate synthetic visuomotor manipulation training data.
- Trained diffusion policies for grasping and manipulating GM parts.
- Built a custom evaluation harness to benchmark grasping and manipulation performance.

Cruise LLC San Francisco, CA | Summer 2023 - 2024
Behaviors PhD Intern (Summer 2024) · Motion Planning & Control PhD Intern (Summer 2023)

- MLE on DriveGPT; autoregressive decoder in a large driving model for NPC vehicle trajectory prediction.
- Kinematic and dynamic metrics for trajectory generation analysis.

AI & Emerging Compute Lab, HP Labs | *HP Inc.* Palo Alto, CA | 2016 - 2020
Robotician & Research Engineer

- Designed and implemented proof of concepts (PoCs) for mobile robots, often with compute-constrained on-board processors utilizing off-board ML servers.
- Developed robotic cleaning solutions for 3D Print, leading a technical team bridging university research (Prof. Phạm Quang-Cường's CRI Group, NTU) to the HP 3D Print business unit's post-processing automation.

- Designed and implemented HP Fused-Deposition-Modeling 3D Printer prototype from first principles.
- Simulated digital twin of hardware with MATLAB and Simulink to conceptualize and prototype multi-input, multi-output control laws for three spatial-axes (xyz), extruder nozzle, and temperature.

PUBLICATIONS

- **J Salfity**, B Anderson, M Pryor “Contract-Grounded Behavior Tree Synthesis via Coding Agents,” (*In submission*). arXiv:2607.12220
- S Swanbeck*, **J Salfity***, J Gunawan, C Van Sice, B Anderson, M Pryor “Deployment Is Not Destiny: Robot Recomposition in the Field with Unseen Software, Hardware, and Compute Payloads,” (*In submission*). arXiv:2608.11063 (**equal contribution*)
- S Wanna, A Luhtaru, **J Salfity**, R Barron, J Moore, C Matuszek, M Pryor “Limited Linguistic Diversity in Embodied AI Datasets,” In *Association for Computational Linguistics (ACL)*, July 2026.
- **J Salfity**, C Van Sice, B Chen, S Swanbeck, B Anderson, M Pryor “No Reboot Required: Real-Time Modular Robot Payload Integration,” In *IEEE International Conference on Ubiquitous Robotics*, July 2025.
- **J Salfity**, S Wanna, M Choi, M Pryor “Temporal and Semantic Evaluation Metrics for Foundation Models in Post-Hoc Analysis of Robotic Subtasks,” Presented at *IEEE International Conference on Ubiquitous Robotics Workshop on Virtual Environment-Based Synthetic Data Generation for Robotic Training*, July 2025.
- S Wanna, A Luhtaru, R Barron, **J Salfity**, J Moore, C Matuszek, M Pryor “Let’s Talk About Language! Investigating Linguistic Diversity in Embodied AI Datasets,” In *IEEE ICRA Safe VLM Workshop*, May 2025.
- Yue Yu, **J Salfity**, D Fridovich-Keil, U Topcu. “Inverse Matrix Games with Unique Quantal Response Equilibrium,” In *IEEE Control Systems Letters*, 2022
- H Nguyen, N Adrian, JLY Yan, **J Salfity**, W Allen, QC Pham. “Development of a robotic system for automated decaking of 3d-printed parts,” In *IEEE International Conference on Robotics and Automation (ICRA)*, May 2020.

PATENTS

- **J Salfity**, W Allen, H Nguyen, N Adrian, JX Yan, QC Pham. 3D Printed Object Cleaning. US12311605B2 (Granted 2025)
- **J Salfity**, D Murphy. Mobile Robots to Generate Reference Maps for Localization. US11703334B2 (Granted 2023)
- **J Salfity**, D Murphy, W Allen. Mobile Robots to Generate Occupancy Maps. US11562524B2 (Granted 2023)
- A Iyer, C Makaya, **J Salfity**. Transmission Rate Modification Based on Predicted Data. US2020/0353193A1 (Filed 2019)
- W Allen, **J Salfity**. Mobile Autonomous Fleet Control. WO2020122953A1 (Filed 2018)
- K Erickson, **J Salfity**, L Zhao. Modules of Three-Dimensional Printers. WO2020046267A1 (Filed 2018)
- **J Salfity**, W Allen, H Horii. Control System for Mobile Robots. WO2019088990A1 (Filed 2017)
- S Stodder, **J Salfity**, M Majette. Correction of Filament Parameters. WO2017086908A1 (Filed 2015)